

```
import java.util.Scanner;
```

```
abstract class Shape {  
    int a, b;  
    abstract void printArea();  
}
```

```
class Rectangle extends Shape {  
    Rectangle(int l, int br) {  
        a = l;  
        b = br;  
    }  

```

```
    void printArea() {  
        System.out.println("Area of Rectangle = " + (a * b));  
    }  
}
```

```
class Triangle extends Shape {  
    Triangle(int base, int height) {  
        a = base;  
        b = height;  
    }  

```

```
void printArea() {  
    System.out.println("Area of Triangle = " + (0.5 * a * b));  
}  
}
```

```
class Circle extends Shape {  
    Circle(int r) {  
        a = r;  
    }  
}
```

```
void printArea() {  
    System.out.println("Area of Circle = " + (3.14 * a * a));  
}  
}
```

```
public class Main {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        System.out.println("Enter Shape (Rectangle/Triangle/Circle):");  
        String shape = sc.next();  
  
        if (shape.equalsIgnoreCase("Rectangle")) {  
            System.out.print("Length:");  
        }  
    }  
}
```

```
int l = sc.nextInt();

System.out.print("Breadth: ");

int b = sc.nextInt();

Shape s = new Rectangle(l, b);

s.printArea();
}

elseif (shape.equalsIgnoreCase("Triangle")) {

    System.out.print("Base: ");

    int base = sc.nextInt();

    System.out.print("Height: ");

    int h = sc.nextInt();

    Shape s = new Triangle(base, h);

    s.printArea();
}

elseif (shape.equalsIgnoreCase("Circle")) {

    System.out.print("Radius: ");

    int r = sc.nextInt();

    Shape s = new Circle(r);

    s.printArea();
}

else {

    System.out.println("Invalid Shape");
}
```

```
        sc.close();  
    }  
}
```